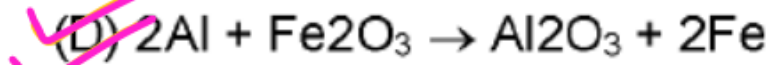
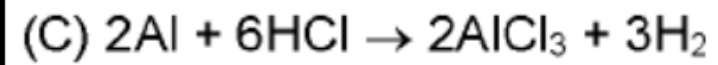
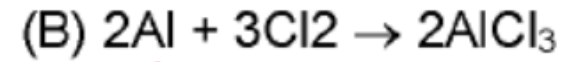
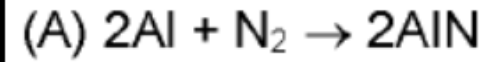
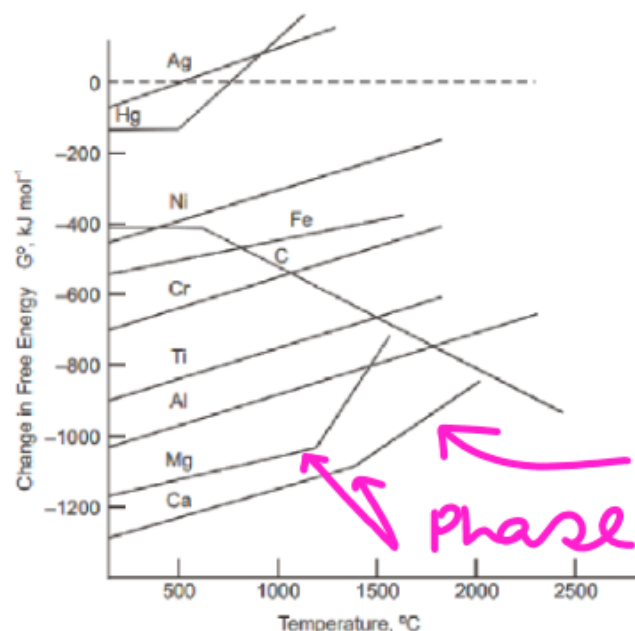


Which of the following reaction forms the basis of Goldschmidt alumino-thermite process ?



Al acting as R-A
for metal oxides like
 Mn_3O_4 , Cr_2O_3 , Fe_2O_3

For a spontaneous reaction, the free energy change must be negative. $\Delta G = \Delta H - T\Delta S$, ΔH is the enthalpy change during the reaction. T is the absolute temperature, and ΔS is the change in entropy during the reaction.



Ellingham diagram.

Which of the following elements can be prepared by heating the oxide below 500°C ?

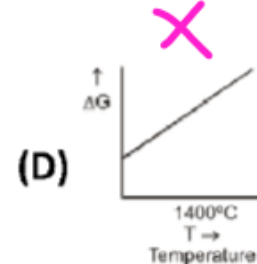
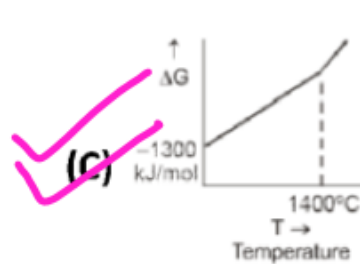
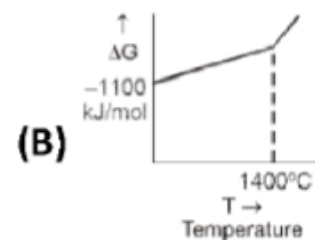
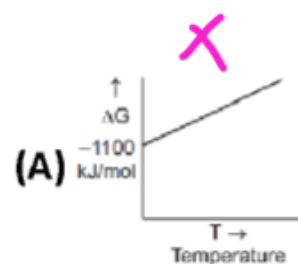
(A) Hg

(B) Mg

(C) Fe

(D) Al

For the conversion of Ca(s) to CaO(s) which of the following represent the ΔG vs. T :



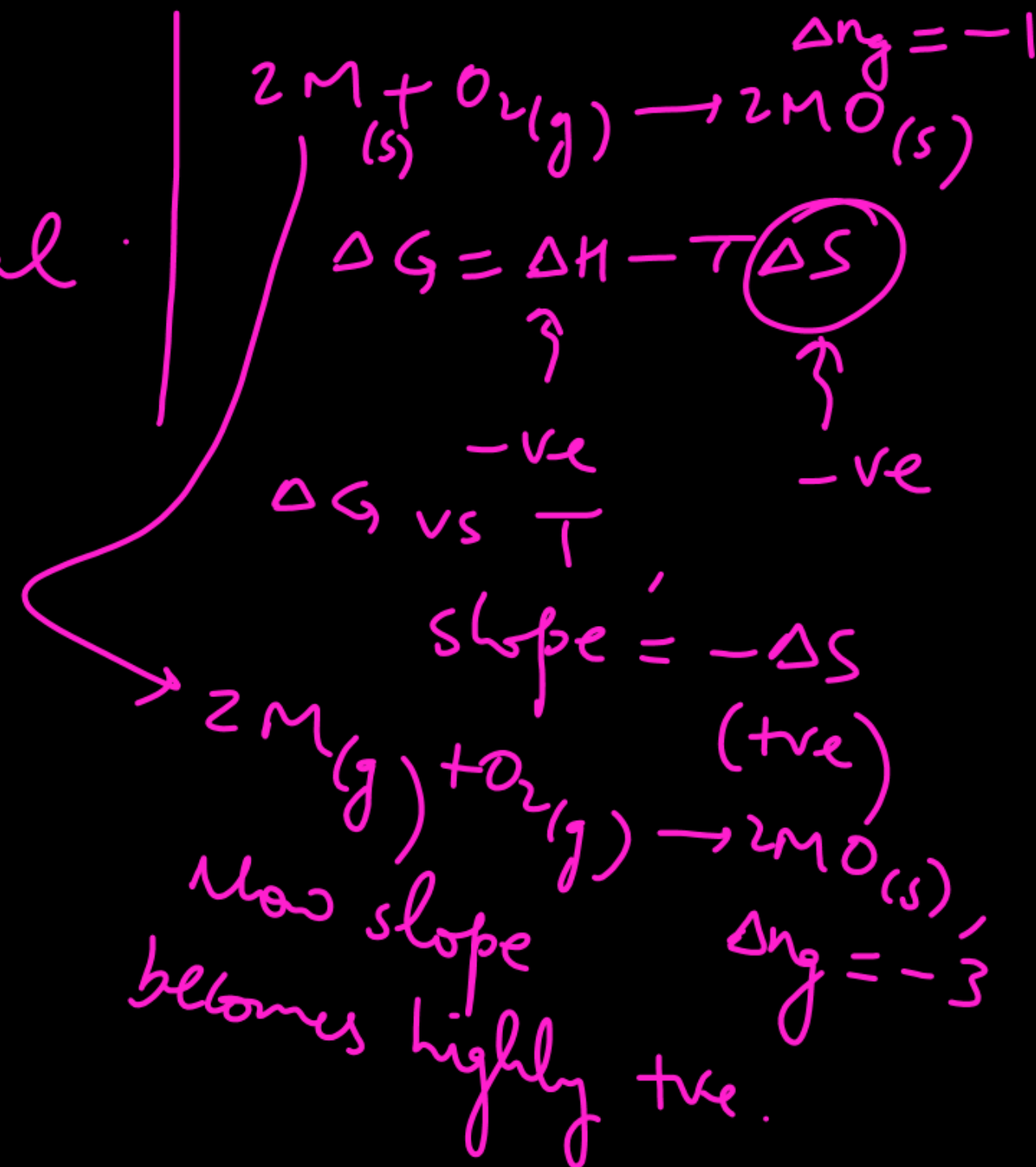
It is seen that free energy change of Hg and Mg for the conversion to oxides the slope of ΔG vs. T has been changed above the boiling points of the given metals because :

(A) above the boiling point of the metal, the entropy is increased.

(B) above the boiling point of the metal, the entropy is decreased in a larger extent.

(C) above the boiling point of the metal, the entropy change is equal to zero.

(D) all of these.



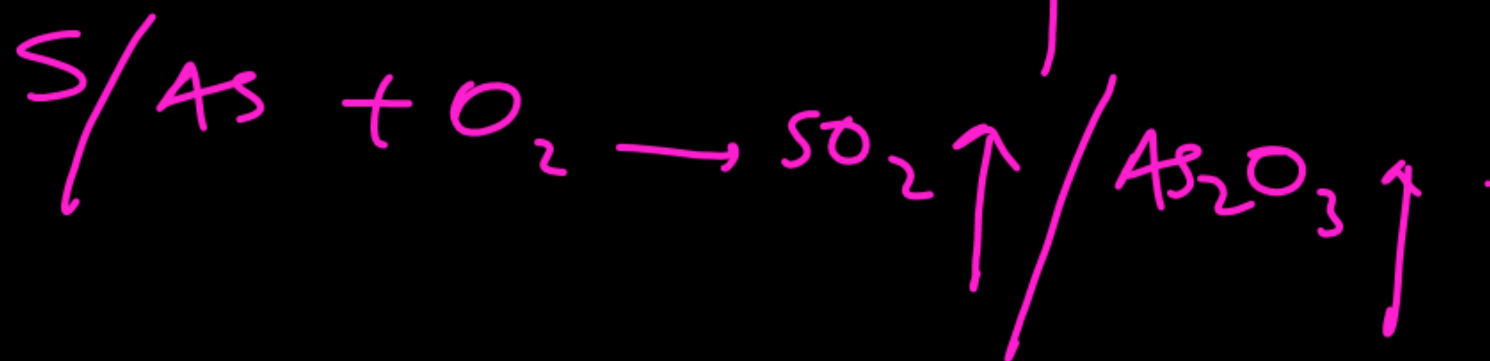
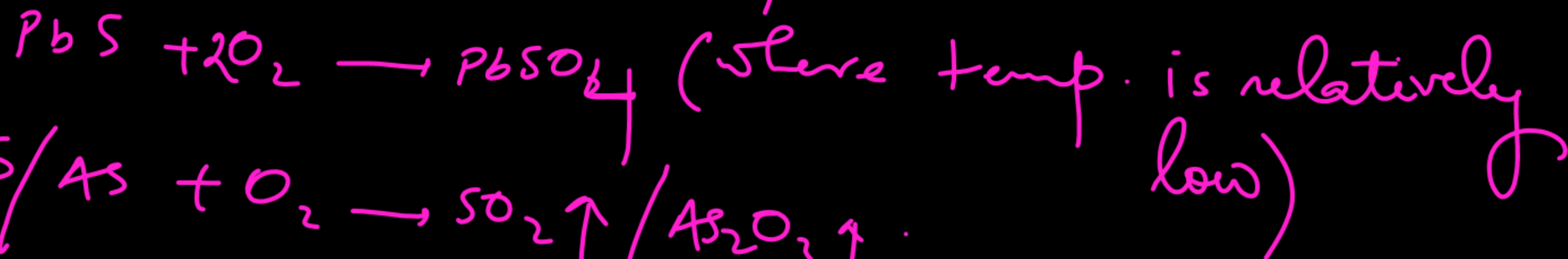
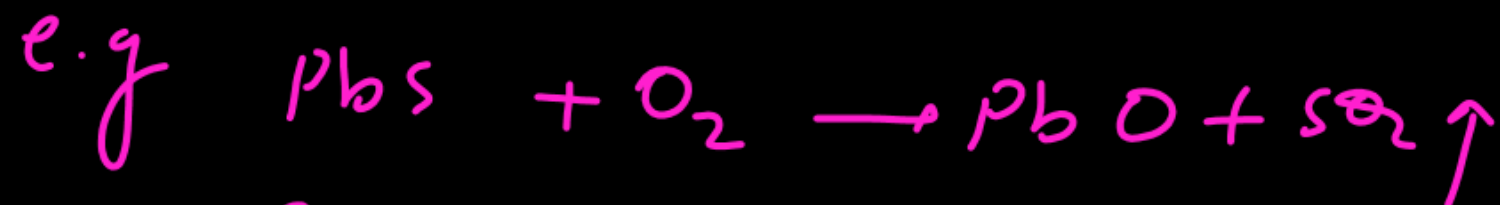
Froth floatation :

- ✓✓ (A) is a physical method of separating mineral from the gangue. tt
- ✓✓ (B) is a method to concentrate the ore depending on the difference in wetability of gangue and the ore.
- ✓✓ (C) is used for the sulphide ores.
- ✓✓ (D) is a method in which impurities sink to the bottom

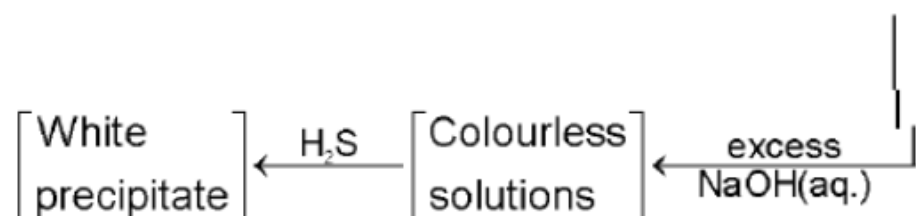
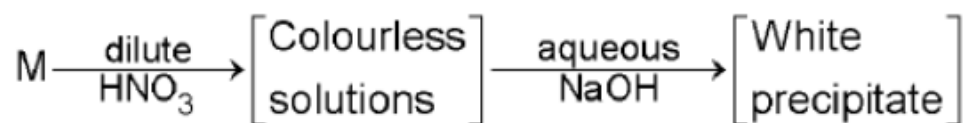
Roasting is carried out to :

- ✓ (A) convert sulphide to oxide and sulphate. ✓ (B) remove water of hydration.
(C) melt the ore. ✓ (D) remove arsenic and sulphur impurities.

X



A metal M and its compound can give the following observable changes in a consequence of reactions



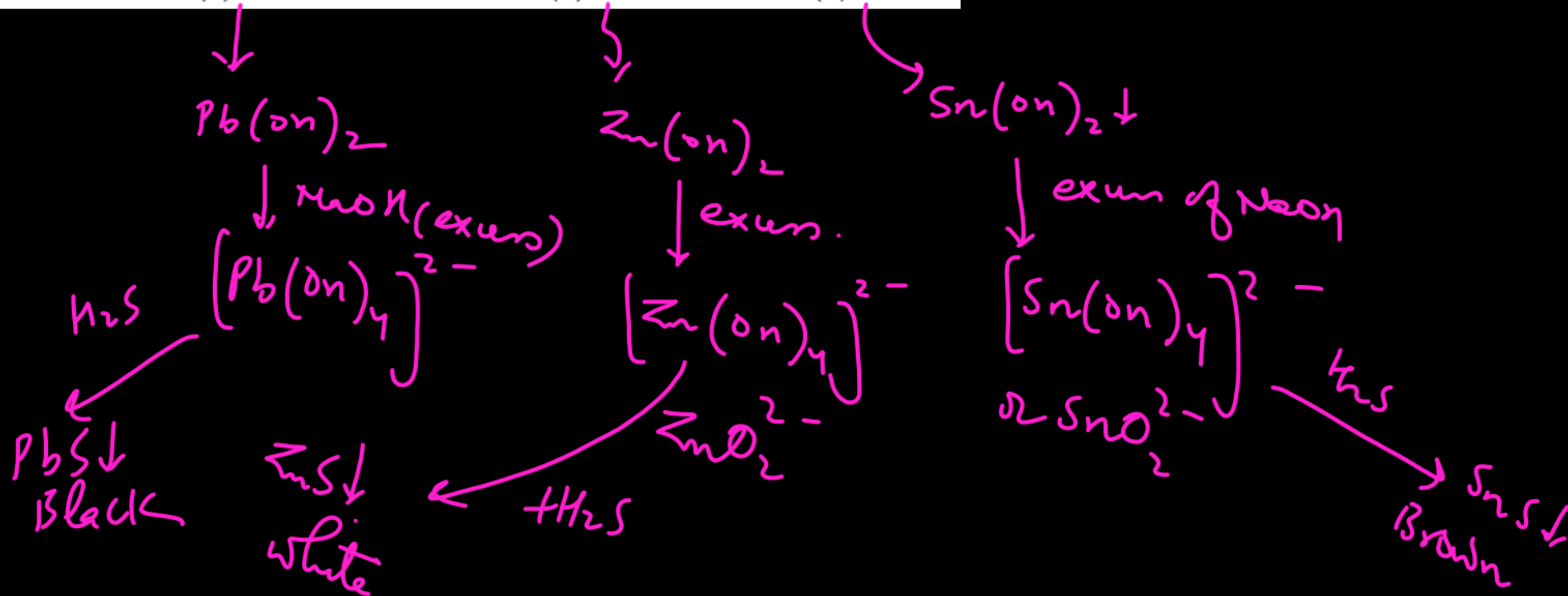
M is:

(A) Mg

(B) Pb

(C) Zn

(D) Sn



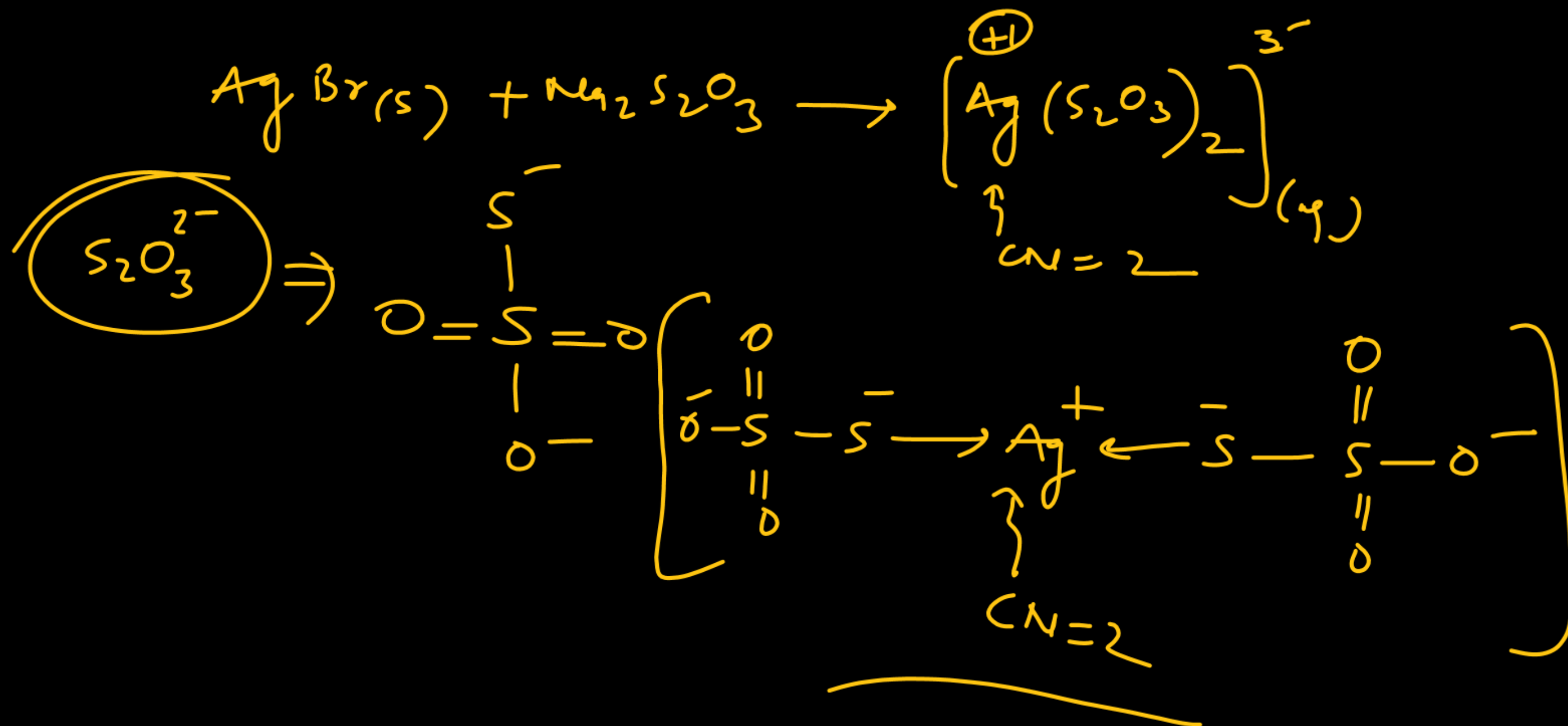
Sodium thiosulphate is used to remove the unexposed AgBr from photographic films by forming a complex. In this complex of silver, the coordination number of silver is

~~(A) 2~~

(B) 4

(C) 6

(D) 8



The false statement about iron (III) hydroxide is that :

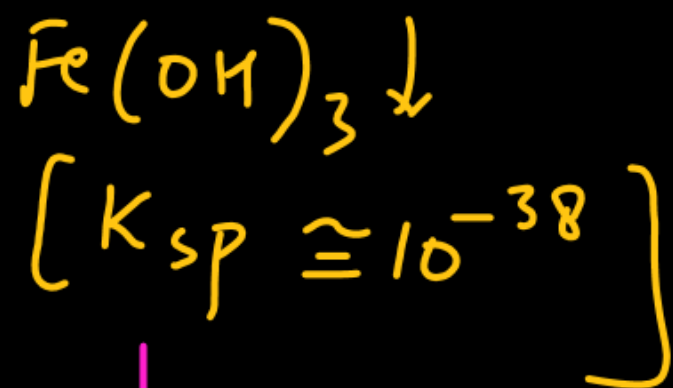
Ans: B, D

(A) it is a weaker base than $\text{Fe}(\text{OH})_2$ ✗

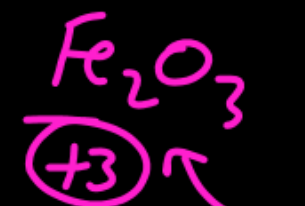
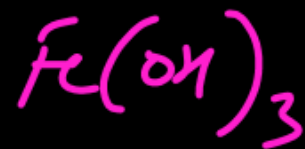
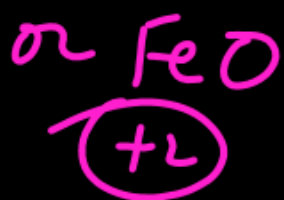
✓ (B) with concentrated KOH , it forms a complex $\text{K}_3[\text{Fe}(\text{OH})_6]$

(C) it gradually loses water and transforms into Fe_2O_3 ✗

✓ (D) it exhibits amphoteric properties with acidic ones predominating



$\downarrow - \text{H}_2\text{O}$



More acidic
or Less basic.

Pyrolusite ore on oxidation with $\text{KClO}_3/\text{KNO}_3$ in basic medium produces dark green coloured compound (A), which on electrolysis produces a purple coloured compound (B). The purple coloured compound can be crystallised to deep purple rhombic prisms. It shows different reactions in different mediums.

Excess of compound (B) on heating with concentrated H_2SO_4 gives an explosive oil (C), which on heating decomposes to give another compound (D) along with oxygen.

On passing CO_2 in the solution of (A), a purple coloured solution along with brown precipitate is observed. Select the correct statement.

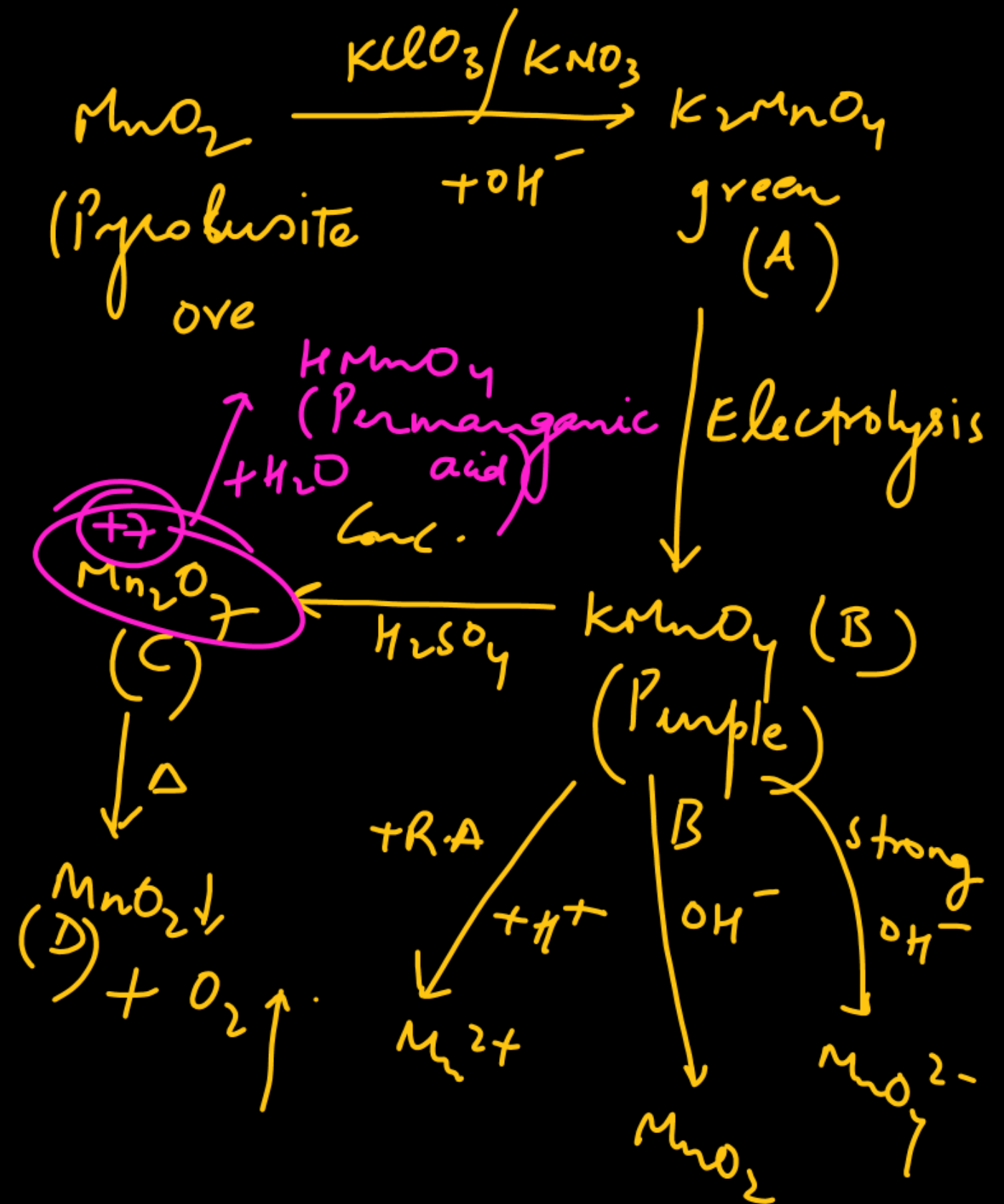
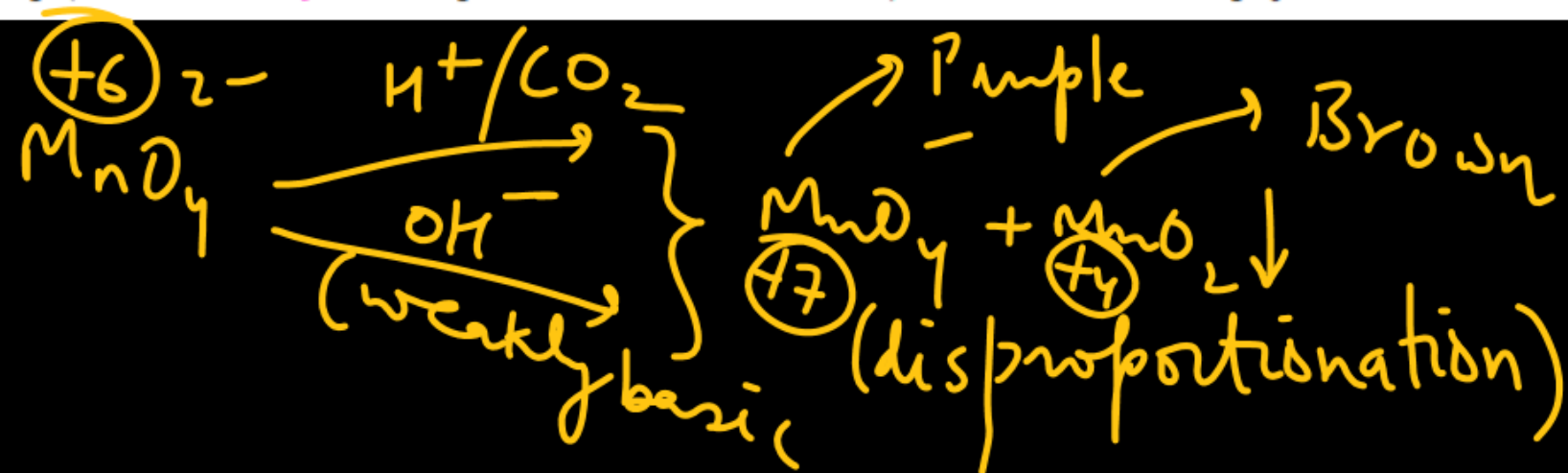
- ✓ (A) This is a disproportionation reaction where oxidation number changes from +6 to +7, +4.
 (B) This is a comproportionation reaction where oxidation number changes from +3 to +2.
 (C) both (A) and (B)
 (D) none

The nature of compound (C) is:

- (A) basic ✓ (B) acidic (C) neutral (D) amphoteric

Identify (D).

- (A) Mn_2O_7 ✓ (B) MnO_2 (C) MnSO_4 (D) Mn_2O_3



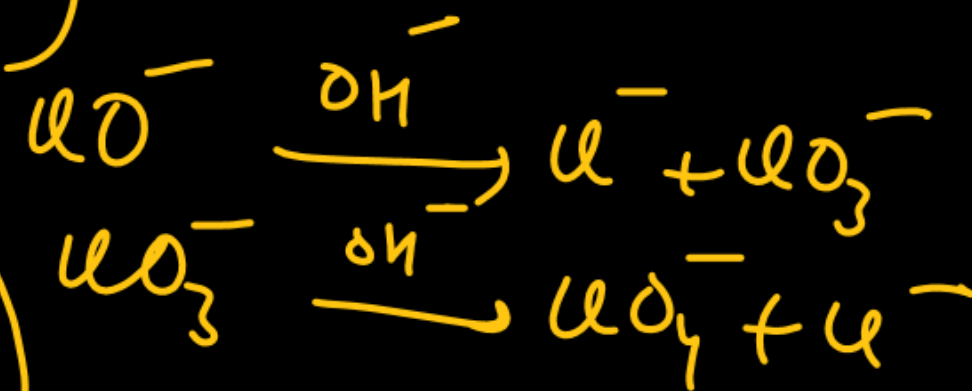
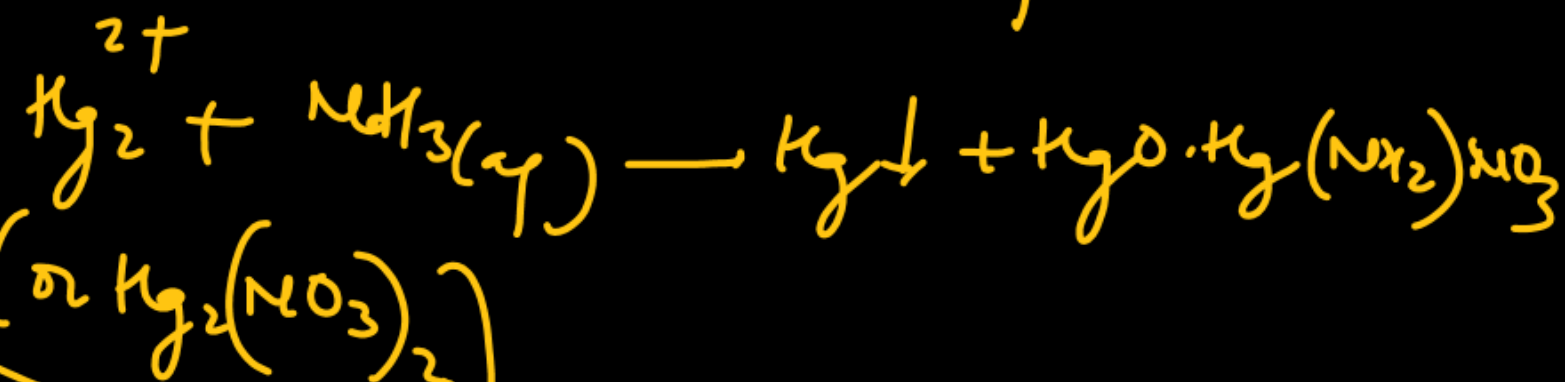
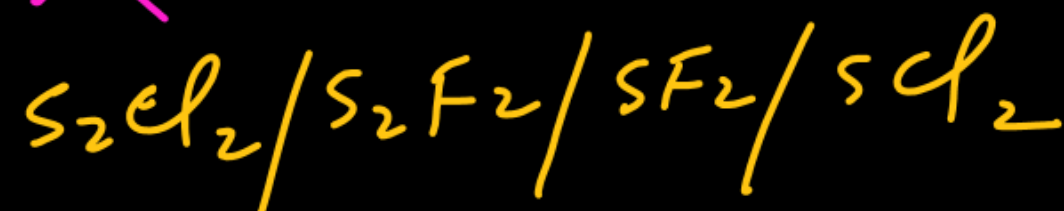
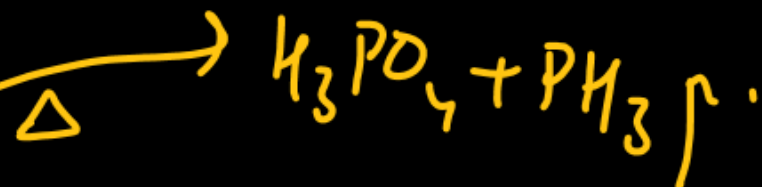
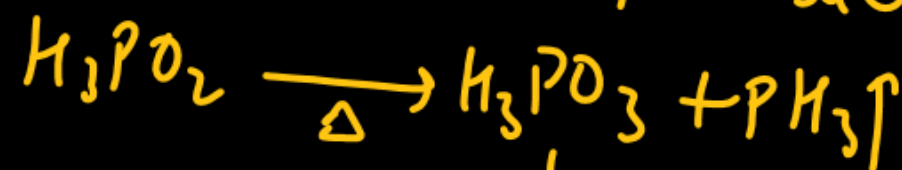
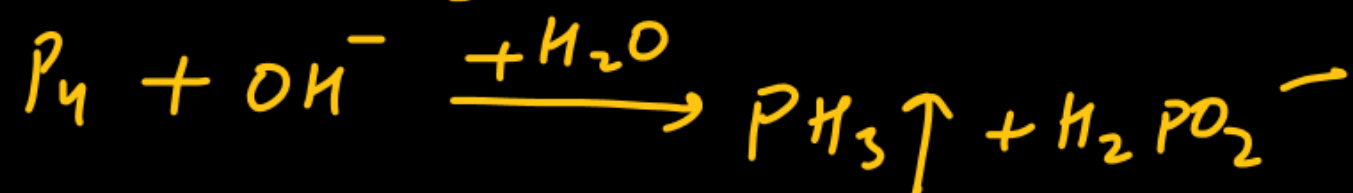
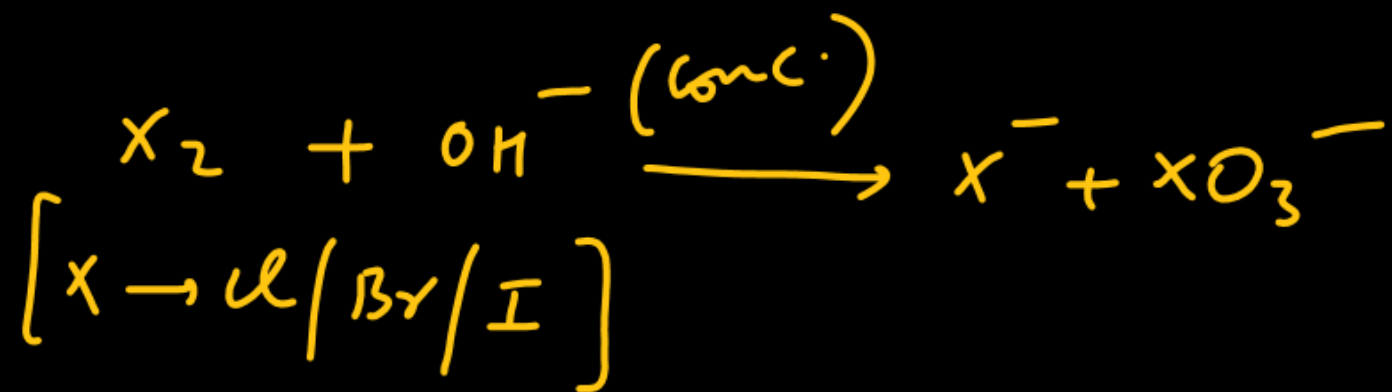
The species that undergoes disproportionation in an alkaline medium are :

(A) Cl_2

(B) MnO_4^{2-}

(C) P_4

(D) ClO_4^-



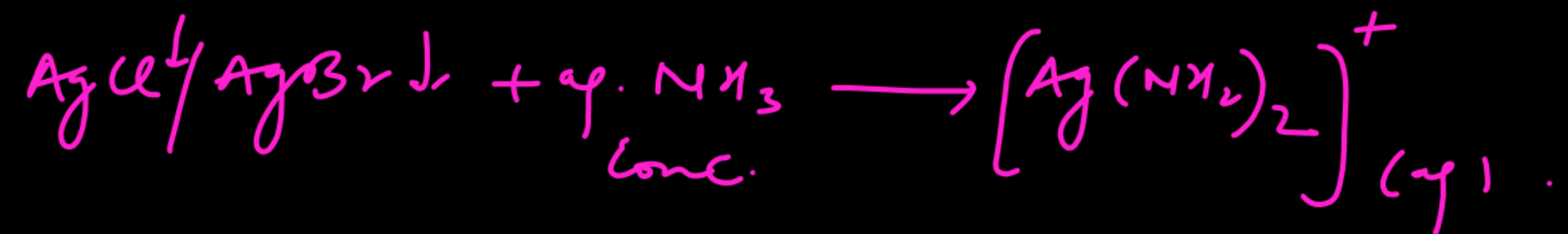
Which is/are insoluble in concentrate NH_3 solution?

(A) AgCl

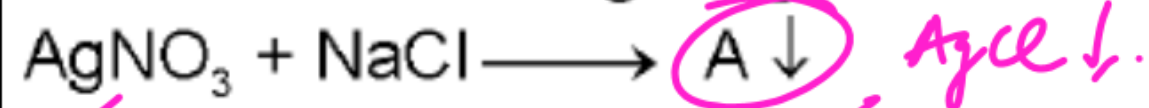
(B) AgBr

~~(C) AgI~~

~~(D) Ag_2S~~



Number of following reagents in which ppt A is soluble is :



✓ (i) aq NH_3

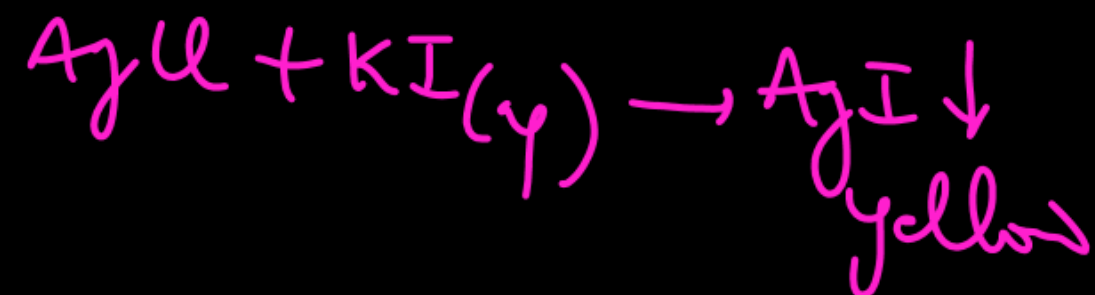
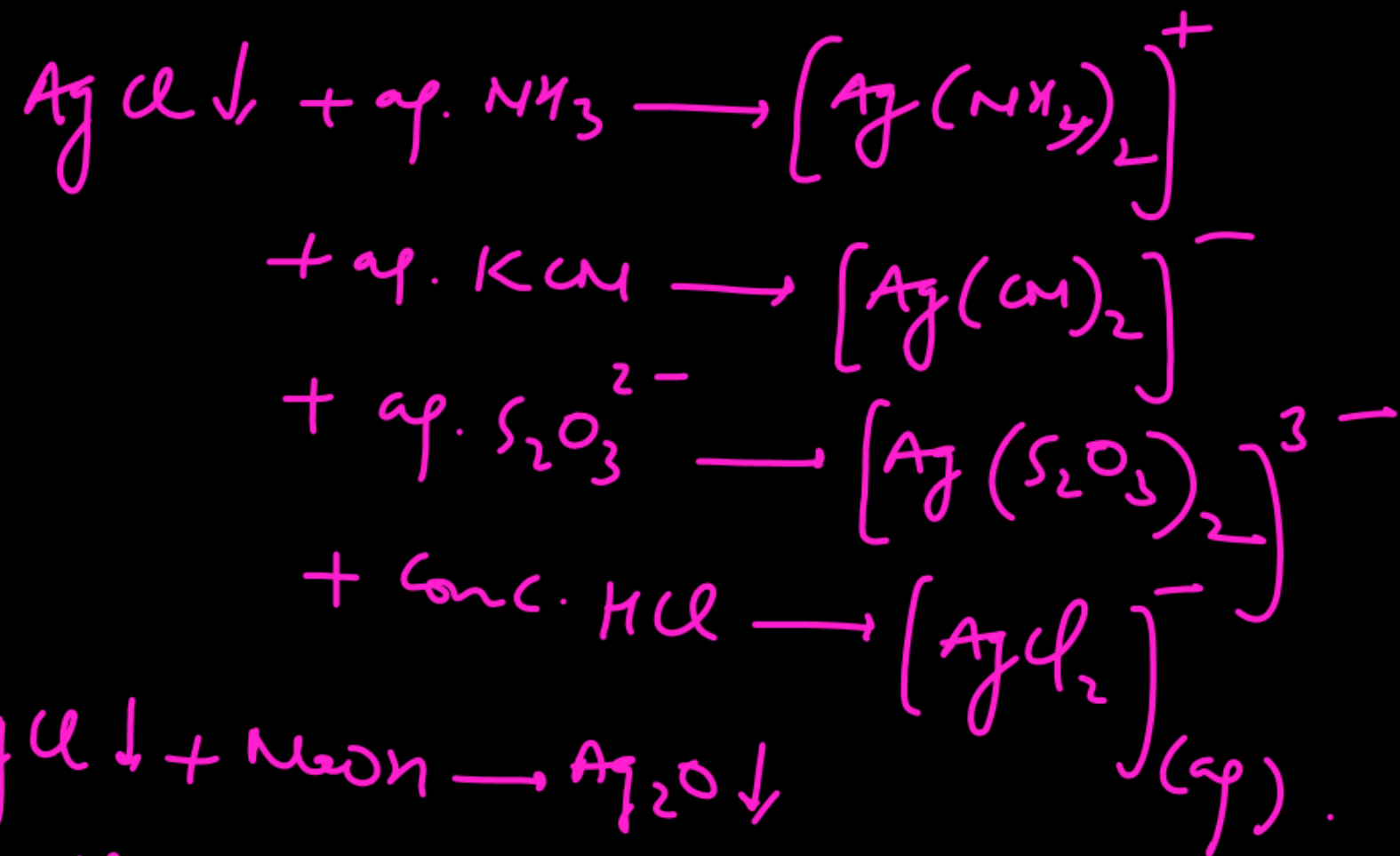
✓ (ii) aq. KCN

✓ (iii) $\text{Na}_2\text{S}_2\text{O}_3$

✓ (iv) Conc. HCl

(v) aq. NaOH X

(iv) aq. KI X

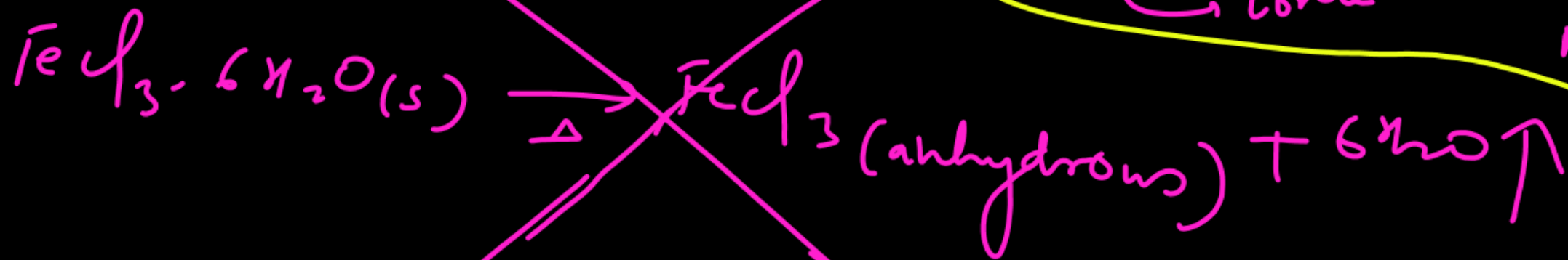


Ans : 4. (aq. NH_3 , aq. KCN,
aq. $\text{Na}_2\text{S}_2\text{O}_3$, aq.
 $\text{Na}_2\text{S}_2\text{O}_3$).

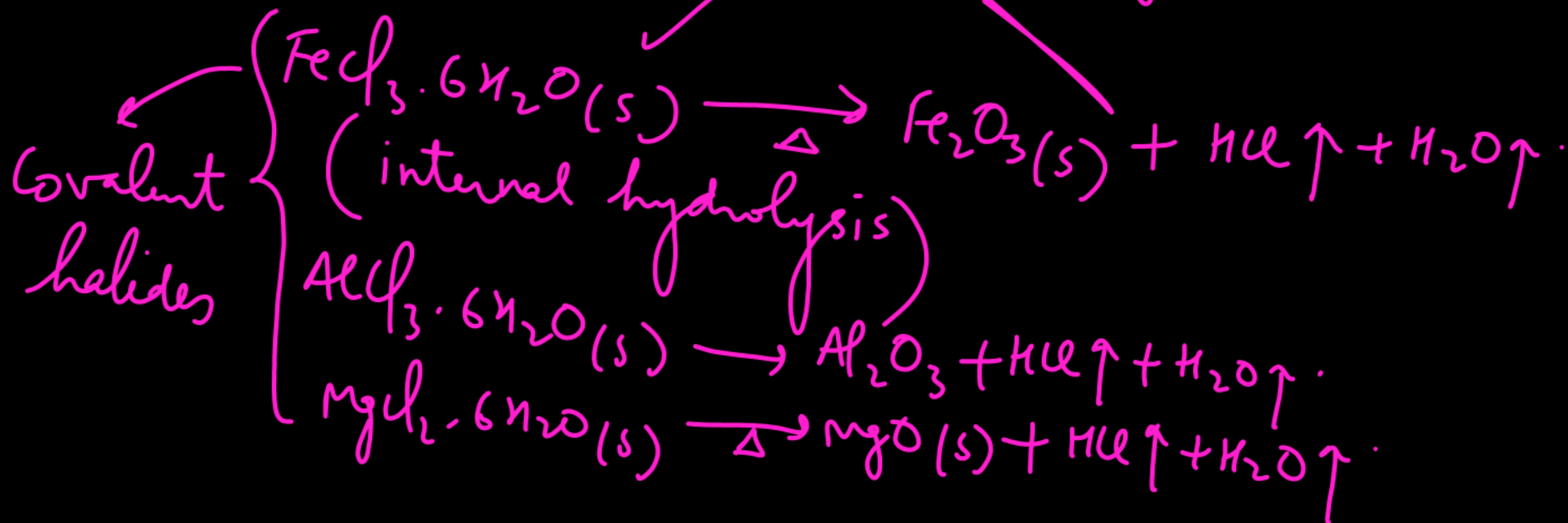
Which of the following reaction(s) is/are used for the preparation of anhydrous FeCl_3 ?

- ☒ (A) $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + 6\text{SOCl}_2 \longrightarrow \text{FeCl}_3 + 12\text{HCl} + 6\text{SO}_2$
☒ (B) $\text{Fe}(\text{OH})_3 \downarrow + 3\text{HCl} \xrightarrow{(\text{aq})} \text{FeCl}_3 + 3\text{H}_2\text{O}$
☒ (C) $2\text{Fe} + 4\text{HCl} (\text{aq.}) + \text{Cl}_2 \xrightarrow{(\text{aq})} 2\text{FeCl}_3 + 2\text{H}_2$
☒ (D) $2\text{Fe} + 3\text{Cl}_2 (\text{dry}) \xrightarrow{(\text{aq})} 2\text{FeCl}_3$

Imp JEE-Adv



$\left. \begin{array}{l} \text{CaCl}_2 \cdot 6\text{H}_2\text{O} \\ \text{SrCl}_2 \cdot 6\text{H}_2\text{O} \\ \text{BaCl}_2 \cdot 6\text{H}_2\text{O} \end{array} \right\}$
 ionic halides
 $\text{MCl}_2 (\text{s}) + 6\text{H}_2\text{O} \uparrow$

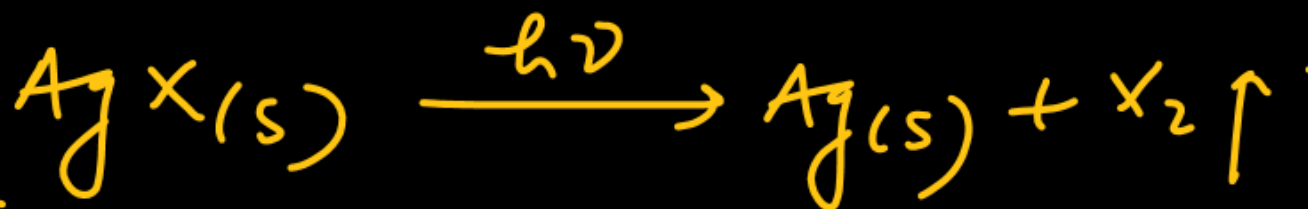
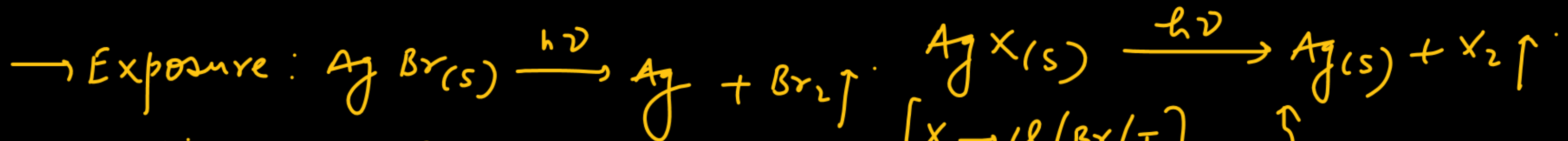


Which of the following chemical reaction(s) is/are involved in developing of photographic plate ?



Ans: A, B

JEE-Adv



grey black

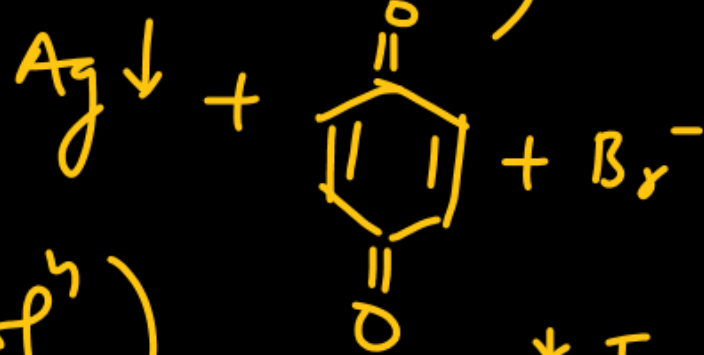
* AgF is not light sensitive.



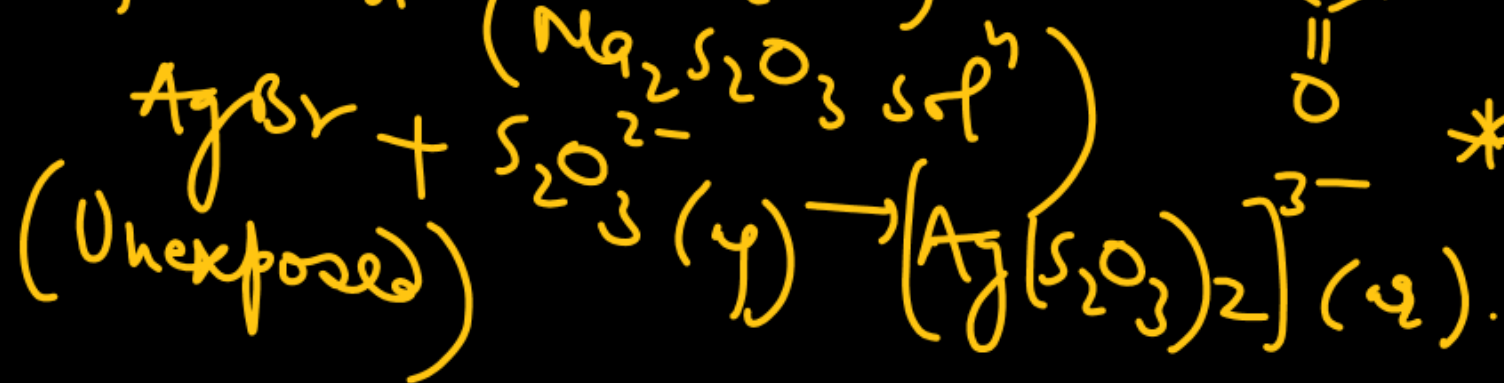
increasing sensitiveness

* In Fast photography, a mix of $AgBr$ & AgI are used.

→ Treating with developer solⁿ

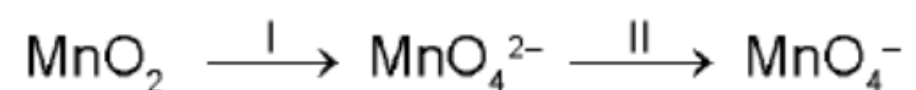


→ fixer solⁿ



(Unexposed)

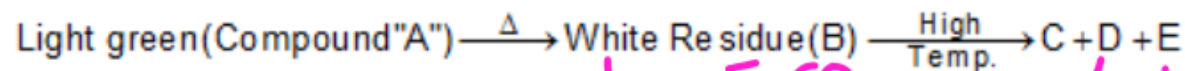
Pyrolusite is MnO_2 used to prepare KMnO_4 . Steps are :



Steps I and II are respectively :

- ✓ (A) fuse with KOH / air, electrolytic oxidation
- ✓ (B) fuse with KOH / KNO_3 , electrolytic oxidation
- ✗ (C) fuse with concentrated HNO_3 / air, electrolytic reduction ✗
- ✗ (D) dissolve in H_2O , oxidation ✗

Ans : A, B.



Observations

- (i) 'D' and 'E' are two acidic gas.
 (ii) 'D' is passed through HgCl_2 solution to give yellow ppt.
 (iii) 'E' is passed through water first and then H_2S is passed, white turbidity is obtained.
 (iv) A is water soluble and addition of HgCl_2 in it, white ppt is obtained but white ppt does not turn into grey on addition of excess solution of 'A'

'D' and 'E' are respectively.

- (A) SO_2 and SO_3 (B) SO_3 and SO_2 (C) SO_2 and CO_2 (D) CO_2 and CO

Yellow ppt in the above observation is :

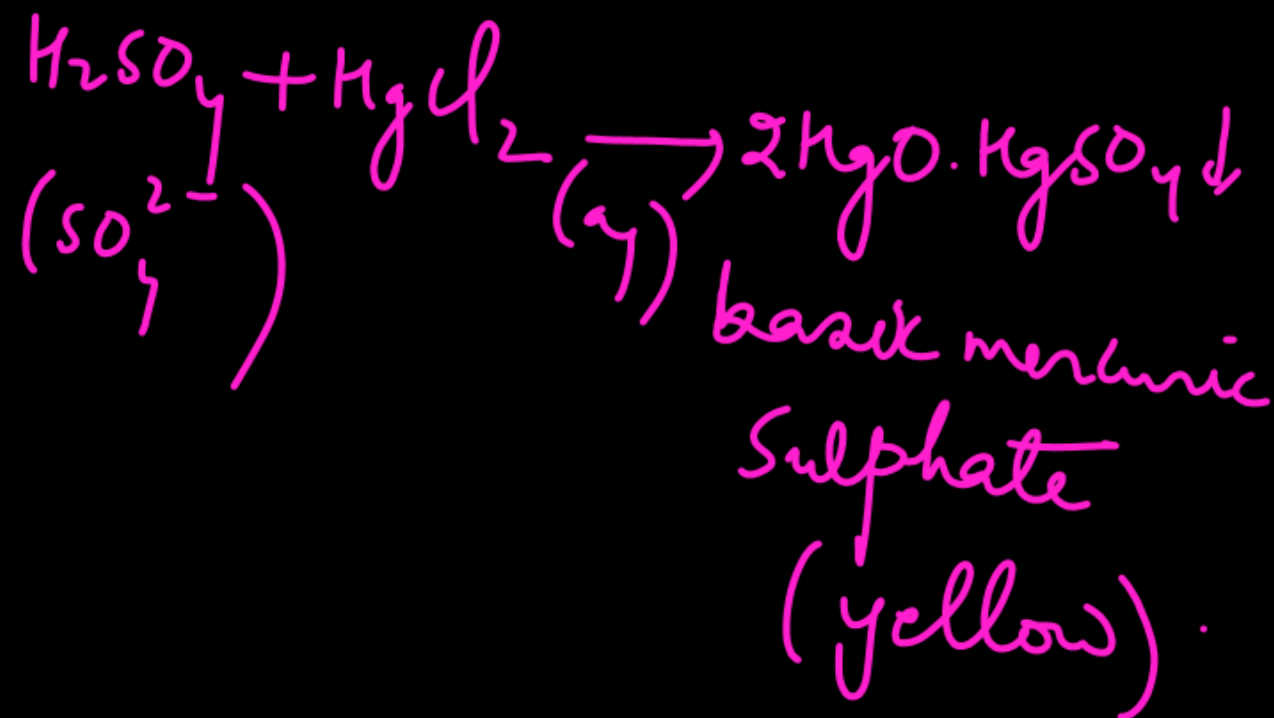
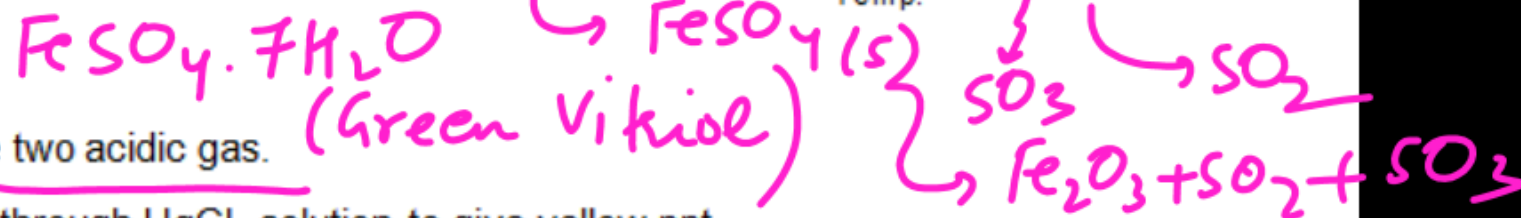
- (A) Mercuric oxide (B) Basic mercury (II) sulphite
 (C) Basic mercury (II) sulphate (D) Mercuric iodide

'C' is soluble in :

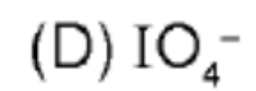
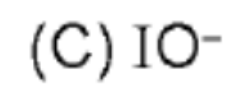
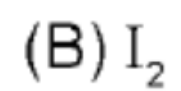
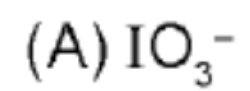
- (A) dil. HCl (B) dil. H_2SO_4 (C) conc. CH_3COOH (D) Boiled conc. HCl

The no. of water of crystallisation in 'A' is :

- (A) 0 (B) 2 (C) 7 (D) 5



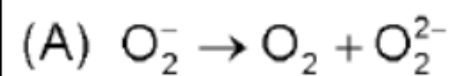
The product of oxidation of I^- with MnO_4^- in alkaline medium is :



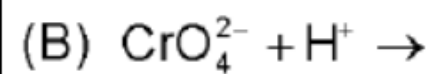
Match the reactions in Column I with nature of the reactions/type of the products in Column II.

Column I

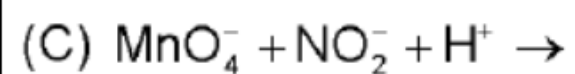
Column II



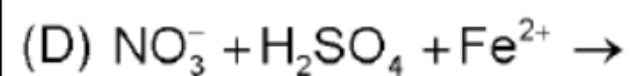
(p) Redox reaction



(q) One of the products has trigonal planar structure



(r) Dimeric bridged tetrahedral metal ion



(s) Disproportionation